Vectorial Model of the Symmetric Three-Phase AC Machine

Extended version: complete derivations and torque sign correction

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Abstract

We present the development of a model using vectorial notation, derived from the equivalence between complex numbers, phasors, and vectors, with heavy subindexing to express in few equations all the variants considered from their original complex-domain form in the reference textbook. This extended version completes three derivations that the original paper stated without development: obtaining the self and mutual inductances from the joint magnetic induction, the algebraic collapse of the three-phase sum in the definition of the flux-linkage space vector, and the full derivation of the torque equation from its original complex form to the final cross-product expression. This last derivation allowed detecting and correcting a sign error in the exponent of the torque equation as it appeared in the original text.

Keywords— Vectorial model, three-phase AC machine, induction motor, cross vector product, flux linkage, mutual inductance

Numbering of the equations. The equations are numbered as in the ACL2 books that verify them [3]. The equations that the books name keep their numbers, for instance (35) or (43). The equations that this extended version adds between two of them carry a letter, for instance (34a)–(34h). Section V lists, for each theorem of the books, the equations it proves.

I. Introduction

The reference three-phase induction motor model uses a complex-domain notation with little generalization of its equations, so it was decided to rewrite it using equations that combine complex notation with the equivalence between complex numbers and phasors or vectors, with heavy subindexing to express in few equations all the variants considered in the original form of the referenced textbook.

Relative to the original publication, this version adds the explicit development of three steps that were there presented as a direct result: the calculation of the self and mutual inductances (Section III.F), the collapse of the sum over the three phases in the definition of the flux-linkage space vector (Section III.H), and the complete derivation of the torque equation (Section III.K), where a sign error relative to the original text is identified and corrected.

II. Complex-Domain AC Motor Model

A. Reference model

Model based on the book by Werner Leonhard, *Control of Electrical Drives*, 2nd and 3rd edition, Springer 1996–2001, Chapter 10, Symmetrical Three-Phase AC Machines [1].

B. Starting assumptions

Two-pole motor, Fig. 1; losses $\to 0$, coil-end effects neglected; balanced Stator and Rotor (S and R) currents, in star or delta three-phase systems; permeability $\mu \to \infty$.

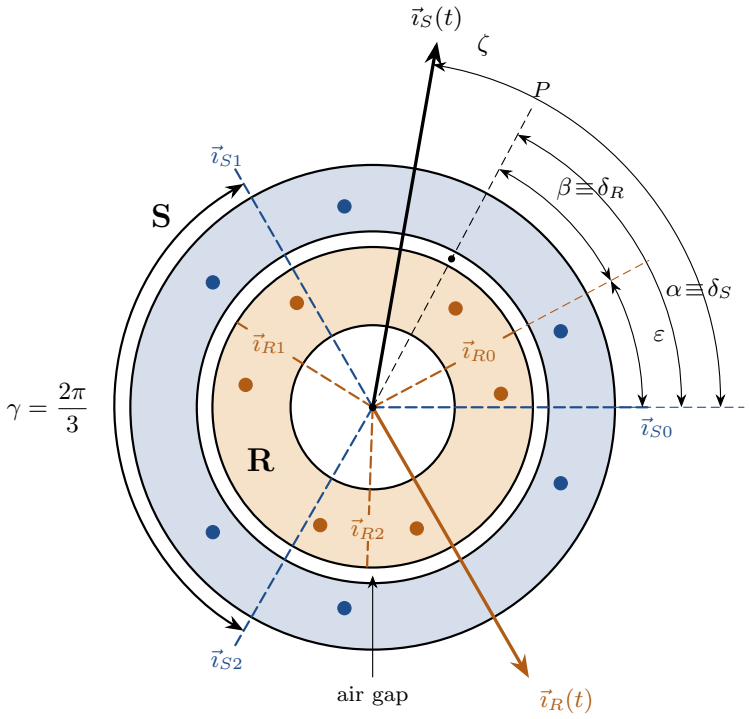


Figure 1: Cross-section of a symmetric three-phase AC machine, with the respective stator and rotor windings distributed and separated by the air gap. Redrawn after Leonhard [1, Fig. 10.2a].

III. Development of the Vectorial AC Motor Model

The progressive development of the vectorial model of the asynchronous AC motor is presented below, starting from the reference complex-domain model.

A. Geometric definitions and mathematical relations

Geometric angle of displacement between coils:

$$\gamma = \tfrac{2}{3}\,\pi \tag{1}$$

Angular coordinate of the stator (S) relative to the central axis of coil 0 of S:

$$\delta_0 \equiv \delta_S \equiv \alpha \tag{2}$$

Angular coordinate of the rotor (R) relative to the central axis of coil 0 of R:

$$\delta_1 \equiv \delta_R \equiv \beta \tag{3}$$

$\varepsilon(t)$: angle of rotation of R relative to the central axis of coil 0 of S. With this we establish the angle relation in the motor:

$$\delta_0 = \delta_1 + \varepsilon(t) \equiv \delta_S = \delta_R + \varepsilon(t) \equiv \alpha = \beta + \varepsilon(t) \tag{4}$$

$$\delta_k = \delta_{k+1} + (-1)^k\,\varepsilon(t) \tag{5}$$

Relation between the product of complex numbers and the scalar (dot) product of vectors [2]:

$$AB^* + A^*B = (a + jb)(c - jd) + (a - jb)(c + jd) = 2(ac + bd) \tag{6}$$

$$A, B \in \mathbb{C} \;\Rightarrow\; AB^* + A^*B = 2\,\vec{A} \bullet \vec{B} \;\Leftarrow\; \vec{A}, \vec{B} \in \mathbb{R}^2 \tag{7}$$

Relation between the product of complex numbers and the cross vector product [2]:

$$AB^* - A^*B = (a+jb)(c-jd) - (a-jb)(c+jd) = j\,2\,(bc-ad) \tag{8}$$

$$A, B \in \mathbb{C} \;\Rightarrow\; AB^* - A^*B = j\,2\,\vec{A} \times \vec{B} \;\Leftarrow\; \vec{A}, \vec{B} \in \mathbb{R}^2 \tag{9}$$

Relation between trigonometric functions, complex numbers, and vectors, using (7):

$$\cos(\delta) = \frac{e^{j\delta} + e^{-j\delta}}{2} \tag{10}$$

$$\cos(\delta-\gamma) = \frac{e^{j(\delta-\gamma)} + e^{-j(\delta-\gamma)}}{2} = \frac{e^{j\delta}e^{-j\gamma} + e^{-j\delta}e^{j\gamma}}{2} \tag{11}$$

$$\cos(\delta-\gamma) = e^{j\delta} \bullet e^{j\gamma} \tag{12}$$

B. Current balance equation

Balance equation for the currents of the electrical system:

$$\sum_n i_{kn}(t) = 0 \tag{13}$$

$$k : \{0,1\} \equiv \{S,R\}\ ; \qquad k+1 = 0 \text{ for } k = 1 \tag{14}$$

k is defined over the ring $\{0,1\}$, so $k+1=0$ for $k=1$.

$$n : \{0,1,2\} \equiv \{R,S,T\} \equiv \{a,b,c\} \tag{15}$$

C. Definition of the stator and rotor current space vector

Definition of the current space vector of S and R (Fig. 2):

$$\vec{\imath}_k(t) = \sum_n i_{kn}(t)\, e^{jn\gamma} \tag{16}$$

Alternative definition of the current space vector of S and R:

$$\vec{\imath}_k(t) = \sum_n i_{kn}(t)\, e^{jn\gamma} = i_k(t)\, e^{j\zeta_k(t)} \tag{17}$$

$\zeta_k(t)$: angle of rotation of $\vec{\imath}_k(t)$ relative to the central axis of coil 0 of k.

Explicit deduction of $\zeta_k(t)$ from (16)–(17). The angle $\zeta_k(t)$ is not an additional unknown of the model: it is deduced by equating the two definitions of the space vector, (16) and (17), and separating real and imaginary parts. Expanding (16) term by term:

$$\vec{\imath}_k(t) = i_{k0}(t)\, e^{j0} + i_{k1}(t)\, e^{j\gamma} + i_{k2}(t)\, e^{j2\gamma} \tag{18}$$

with $\gamma = 2\pi/3$, so the two complex roots involved are:

$$e^{j\gamma} = -\frac{1}{2} + j\frac{\sqrt{3}}{2}\ , \qquad e^{j2\gamma} = -\frac{1}{2} - j\frac{\sqrt{3}}{2} \tag{19}$$

Separating the real and imaginary parts of $\vec{\imath}_k(t)$:

$$\mathrm{Re}\{\vec{\imath}_k(t)\} = i_{k0}(t) - \tfrac{1}{2}\, i_{k1}(t) - \tfrac{1}{2}\, i_{k2}(t) \tag{20}$$

$$\mathrm{Im}\{\vec{\imath}_k(t)\} = \frac{\sqrt{3}}{2}\,[\,i_{k1}(t) - i_{k2}(t)\,] \tag{21}$$

The current balance condition (13) implies that the sum of phases 1 and 2 is equal and opposite to phase 0:

$$i_{k1}(t) + i_{k2}(t) = -\,i_{k0}(t) \tag{22}$$

with which the real part simplifies to:

$$\mathrm{Re}\{\vec{\imath}_k(t)\} = i_{k0}(t) - \tfrac{1}{2}(-i_{k0}(t)) = \tfrac{3}{2}\,i_{k0}(t) \tag{23}$$

resulting in the form equivalent to the Clarke (α-β) transform, written as a complex number:

$$\vec{\imath}_k(t) = \tfrac{3}{2}\,i_{k0}(t) + j\,\frac{\sqrt{3}}{2}\,[\,i_{k1}(t) - i_{k2}(t)\,] \tag{24}$$

Equating this expression with the polar form (17) —$i_k(t)e^{j\zeta_k(t)} = i_k(t)\cos\zeta_k(t) + j\,i_k(t)\sin\zeta_k(t)$— component by component:

$$i_k(t)\cos\zeta_k(t) = \tfrac{3}{2}\,i_{k0}(t)\ , \qquad i_k(t)\sin\zeta_k(t) = \frac{\sqrt{3}}{2}\,[\,i_{k1}(t) - i_{k2}(t)\,] \tag{25}$$

Dividing both expressions cancels $i_k(t)$, and $\zeta_k(t)$ is obtained explicitly as a function of the three phase currents (atan2 is used instead of plain arctan to preserve the correct quadrant, since i_{k0} can be negative):

$$\zeta_k(t) = \mathrm{atan2}\Big(\frac{\sqrt{3}}{2}\,[\,i_{k1}(t) - i_{k2}(t)\,],\ \tfrac{3}{2}\,i_{k0}(t)\Big) \tag{26}$$

And along the way, dividing instead by the cosine or the sine and summing the squares, the magnitude is also determined:

$$i_k(t) = \frac{3}{2}\sqrt{i_{k0}^2(t) + \tfrac{1}{3}\,[\,i_{k1}(t) - i_{k2}(t)\,]^2} \tag{27}$$

Alternative physical verification. From (28h), $\Theta_k(\delta_k, t) = N_k\,i_k(t)\cos[\zeta_k(t) - \delta_k]$ is a traveling cosine wave in the spatial variable δ_k. Its maximum —the crest of the mmf wave at instant t— occurs exactly where the derivative with respect to δ_k vanishes:

$$\frac{\partial\Theta_k}{\partial\delta_k} = 0 \ \Rightarrow\ \sin[\zeta_k(t) - \delta_k] = 0 \ \Rightarrow\ \delta_k = \zeta_k(t) \tag{28}$$

That is: $\zeta_k(t)$ is, at each instant, the angular position where the mmf wave of winding k reaches its maximum value along the air gap — the same geometric interpretation shown by Fig. 2 as the angle of the resultant vector from the geometric sum of the three phase currents. Both paths, the algebraic one via the Clarke transform and the physical one via the maximum of the traveling wave, are mutually consistent and confirm that $\zeta_k(t)$ is fully determined by $i_{k0}(t)$, $i_{k1}(t)$, and $i_{k2}(t)$.

D. Calculation of the stator and rotor mmf

Calculation of the magnetomotive force of the stator (S) and rotor (R). This forms a traveling wave, as shown in Fig. 3.

$$\Theta_k(\delta_k, t) = N_k \sum_n i_{kn}(t)\,\vartheta(\delta_k - n\gamma) \tag{28a}$$

$$-1 \le \vartheta(\delta_k - n\gamma) \le +1 \tag{28b}$$

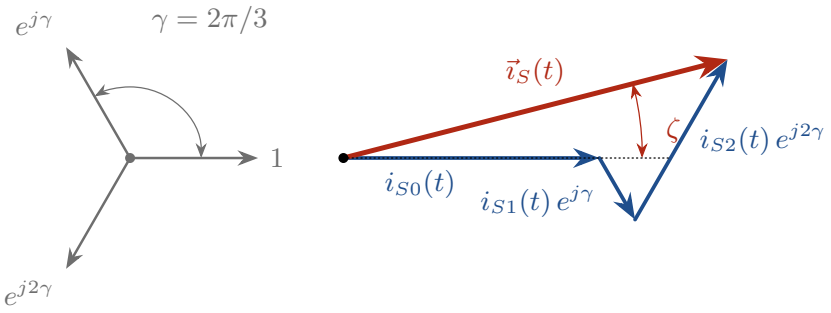


Figure 2: Current space vector as the geometric sum of the stator winding currents, for the instantaneous balanced values $(i_{S0}, i_{S1}, i_{S2}) = (1.8, -0.5, -1.3)$. Redrawn after Leonhard [1, Fig. 10.3a].

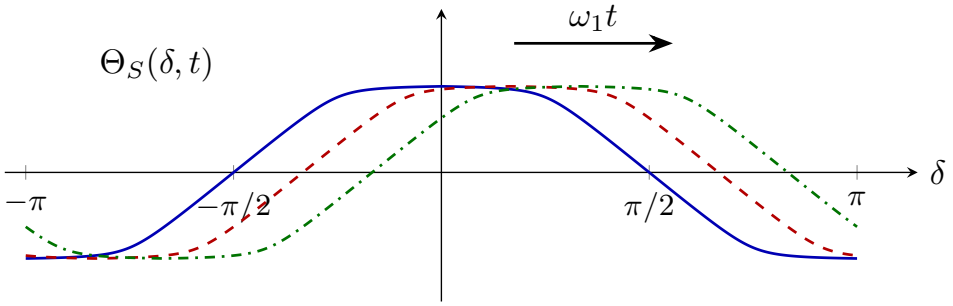


Figure 3: Traveling magnetomotive-force wave of the stator, resulting from combining the contributions of its individual windings. Redrawn after Leonhard [1, Fig. 10.2b].

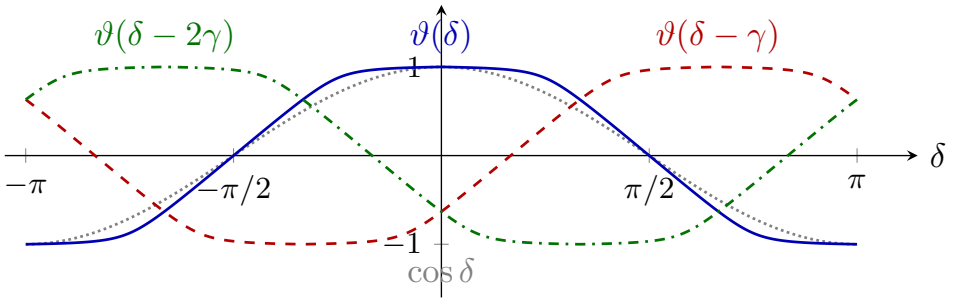


Figure 4: Distribution functions of the magnetomotive force of the stator windings. Redrawn after Leonhard [1, Fig. 10.2c].

$\vartheta(\delta_k - n\gamma)$: spatial distribution function of the magnetomotive force of winding n, as a function of the stator or rotor measurement angle δ_k. See Fig. 4.

A harmonic-elimination approximation is made:

$$\vartheta(\delta_k - n\gamma) \cong \cos(\delta_k - n\gamma) \tag{28c}$$

Applying the previous relations and definitions to the mmf (4) of S and R:

$$\Theta_k(\delta_k, t) = N_k \sum_n i_{kn}(t) \cos(\delta_k - n\gamma) \tag{28d}$$

$$\Theta_k(\delta_k, t) = N_k \sum_n i_{kn}(t) \left(e^{j\delta_k} \bullet e^{jn\gamma}\right) \tag{28e}$$

$$\Theta_k(\delta_k, t) = N_k\, e^{j\delta_k} \bullet \Big[\sum_n i_{kn}(t)\, e^{jn\gamma}\Big] \tag{28f}$$

$$\Theta_k(\delta_k, t) = N_k\, \vec{i}_k(t) \bullet e^{j\delta_k} \tag{28g}$$

Alternative definition of $\Theta_k(t)$:

$$\Theta_k(\delta_k, t) = N_k\, i_k(t) \cos[\zeta_k(t) - \delta_k] \tag{28h}$$

Given that by (12):

$$\cos(\zeta - \delta) = e^{j\zeta} \bullet e^{j\delta} \tag{28i}$$

We obtain:

$$\Theta_k(\delta_k, t) = N_k\, i_k(t) \left[e^{j\zeta_k(t)} \bullet e^{j\delta_k}\right] \tag{28j}$$

And the total mmf:

$$\Theta(t) = \sum_k \Theta_k(\delta_k, t) \tag{28k}$$

E. Calculation of the stator and rotor magnetic induction

Magnetic induction or magnetic flux density in S or R due to the joint mmfs, using the angle relation between stator and rotor (5):

$$B_k(\delta_k, \varepsilon(t), t) = \frac{1}{2h}\, \mu_0 \Big\{(1 + \sigma_k)\, \Theta_k(\delta_k, t) + \Theta_{k+1}\big[\delta_k - (-1)^k \varepsilon(t),\, t\big]\Big\} \tag{29}$$

Equivalent form as a function of each member's own angles:

$$B_k(\delta_k, \delta_{k+1}, t) = \frac{1}{2h}\, \mu_0 \Big[(1 + \sigma_k)\, \Theta_k(\delta_k, t) + \Theta_{k+1}(\delta_{k+1}, t)\Big] \tag{29b}$$

where h is the air-gap thickness and σ_k are the magnetic losses of S or R:

$$0 \le \sigma_k < 1 \tag{29c}$$

F. Self and mutual inductances — complete derivation

The original paper presents this result as a direct consequence of "integrating the inductive couplings or flux linkages," without developing it. That step is completed here. The method is the usual "winding function" one: the flux linked by phase n of member k is obtained by integrating the induction B_k over the air gap, weighted by the distribution function (or "pickup function") of that same coil. For a two-pole distributed winding, this pickup function has amplitude $N_k/2$ —half

the amplitude N_k used in the mmf (28a), the standard distinction between the "turns function" (for the mmf) and the "winding function" (for the flux linkage)— so that:

$$\psi_{kn}(t) = \frac{N_k}{2}\, r\, \ell \int_0^{2\pi} B_k(\delta_k, \varepsilon(t), t)\, \cos(\delta_k - n\gamma)\, d\delta_k \tag{29d}$$

Using the product-to-sum trigonometric identity:

$$\int_0^{2\pi} \cos(\delta_k - m\gamma)\, \cos(\delta_k - n\gamma)\, d\delta_k = \pi \cos[(m-n)\gamma] \tag{29e}$$

Self term. Substituting (29b) for B_k and applying the previous identity to the $(1+\sigma_k)\Theta_k$ part, together with the balanced-current condition (13), gives the identity:

$$\sum_m i_{km}(t)\, \cos[(m-n)\gamma] = \tfrac{3}{2}\, i_{kn}(t) \tag{29f}$$

with which the self term of the flux linkage becomes:

$$\tfrac{1}{3} L_k = \frac{(1+\sigma_k)\, N_k^2\, \ell\, r\, \pi\, \mu_0}{8h} \tag{30}$$

which reproduces exactly the result of the original paper.

Mutual term. Similarly, with $\varphi \equiv (-1)^k \varepsilon(t)$, the same product-to-sum integral over the contribution of member $k+1$ gives:

$$\sum_p i_{k+1,p}(t)\, \cos[\varphi - (n-p)\gamma] = [\vec{i}_{k+1}(t)\, e^{j\varphi}] \bullet e^{jn\gamma}, \qquad \varphi \equiv (-1)^k \varepsilon(t) \tag{30a}$$

which —after reconstituting the current space vector of member $k+1$ and comparing with the expected form of equation (32) below— determines the mutual inductance:

$$\tfrac{1}{3} M = \frac{N_k\, N_{k+1}\, \ell\, r\, \pi\, \mu_0}{8h} \tag{31}$$

Both results (30) and (31) share the same denominator $8h$, and their ratio $L_k/M = (1+\sigma_k)N_k/N_{k+1}$ is identical to the one obtained by Leonhard's classical scalar path, which confirms the consistency of the vectorial model with the original complex model.

G. Calculation of the stator and rotor inductive couplings

Using (30) and (31), the inductive couplings of stator and rotor take the following vectorial form:

$$\psi_{kn}(t) = \tfrac{2}{3}\Big\{ L_k\, \vec{i}_k(t) \bullet e^{jn\gamma} + M\, \vec{i}_{k+1}(t) \bullet e^{j[n\gamma - (-1)^k \varepsilon(t)]} \Big\} \tag{32}$$

H. Definition of the inductive-coupling space vector — complete derivation

Definition of the inductive-coupling space vector of S and R:

$$\vec{\psi}_k(t) = \sum_n \psi_{kn}(t)\, e^{jn\gamma} \tag{33}$$

Substituting (32) into (33):

$$\vec{\psi}_k(t) = \sum_n \tfrac{2}{3}\Big\{ L_k\, \vec{i}_k(t) \bullet e^{jn\gamma} + M\, \vec{i}_{k+1}(t) \bullet e^{j[n\gamma - (-1)^k \varepsilon(t)]} \Big\}\, e^{jn\gamma} \tag{34}$$

The original paper summarizes this step as "after some algebraic elaboration." That elaboration is completed here by means of a three-phase collapse lemma, a direct consequence of the system's 120° symmetry (the same kind of cancellation that underlies the current balance (13)–(15)):

$$\sum_{n=0}^{2} \tfrac{2}{3}\,(A \bullet e^{jn\gamma})\, e^{jn\gamma} = A \qquad \left(\gamma = \frac{2\pi}{3}\right) \tag{34a}$$

Proof of the lemma. For any $A \in \mathbb{C}$ constant in n, the dot product can be written as a real part:

$$A \bullet e^{jn\gamma} = \mathrm{Re}\big(A\, e^{-jn\gamma}\big) = \frac{A\, e^{-jn\gamma} + A^{*} e^{jn\gamma}}{2} \tag{34b}$$

multiplying by $(2/3)e^{jn\gamma}$:

$$\tfrac{2}{3}(A \bullet e^{jn\gamma})\, e^{jn\gamma} = \tfrac{1}{3}(A + A^{*} e^{j2n\gamma}) \tag{34c}$$

and summing over $n = 0, 1, 2$:

$$\sum_{n=0}^{2} \tfrac{1}{3}(A + A^{*} e^{j2n\gamma}) = A + \frac{A^{*}}{3} \sum_{n=0}^{2} e^{j2n\gamma} \tag{34d}$$

With $\gamma = 2\pi/3$, the exponent $2n\gamma$ runs over the three cube roots of unity, whose sum is identically zero:

$$\sum_{n=0}^{2} e^{jn\frac{4\pi}{3}} = 1 + e^{j\frac{4\pi}{3}} + e^{j\frac{8\pi}{3}} = 0 \tag{34e}$$

so the second term vanishes and the whole sum collapses exactly to A, proving the lemma above.

Application to the self term of (34), with $A = L_k \vec{\imath}_k(t)$ (constant in n):

$$\sum_{n} \tfrac{2}{3}\big[L_k\, \vec{\imath}_k(t) \bullet e^{jn\gamma}\big] e^{jn\gamma} = L_k\, \vec{\imath}_k(t) \tag{34f}$$

Application to the mutual term. First bringing the exponent to the pure $e^{jn\gamma}$ form via a phase shift φ:

$$B \bullet e^{j(n\gamma - \varphi)} = \big(B\, e^{j\varphi}\big) \bullet e^{jn\gamma} \tag{34g}$$

with $B = \vec{\imath}_{k+1}(t)$ and $\varphi = (-1)^k \varepsilon(t)$, the term is also constant in n and the lemma applies directly:

$$\sum_{n} \tfrac{2}{3}\Big[M\, \vec{\imath}_{k+1}(t)\, e^{j(-1)^k \varepsilon(t)} \bullet e^{jn\gamma}\Big] e^{jn\gamma} = M\, \vec{\imath}_{k+1}(t)\, e^{j(-1)^k \varepsilon(t)} \tag{34h}$$

Summing both results gives exactly the result stated in the original paper:

$$\vec{\psi}_k(t) = L_k\, \vec{\imath}_k(t) + M\, \vec{\imath}_{k+1}(t)\, e^{j(-1)^k \varepsilon(t)} \tag{35}$$

The trigonometric sum over the three phases cancels completely thanks to the 120° geometric phase shift between windings: the flux-linkage space vector is expressed as the superposition of the self flux with the mutual flux coupled from the other winding, rotated by the corresponding mechanical angle.

Angular speed of the mmf of $k \in \{S, R\}$:

$$\omega_k(t) = \dot{\zeta}_k(t) \tag{36}$$

Angular speed of the motor of $k \in \{S, R\}$:

$$\omega_M(t) = \dot{\varepsilon}(t) \tag{37}$$

Note: (37) is not obtained algebraically from (36), but by analogical extension of the same definition —angular speed as the time derivative of a rotation angle— to the mechanical angle $\varepsilon(t)$ already defined in (4)–(5), instead of the electrical angle $\zeta_k(t)$. Both angles describe a rotation relative to a fixed reference axis, but one is electrical and the other mechanical; they are not in general equal or proportional to each other without additional information (hence the slip in the induction machine).

I. Calculation of the stator and rotor voltages

Analogously, what was done can be applied to the voltage equation of S and R:

$$u_{kn}(t) = R_k\, i_{kn}(t) + \dot{\psi}_{kn}(t) \tag{38}$$

J. Definition of the stator and rotor voltage space vector

Definition of the voltage space vector of S and R:

$$\vec{u}_k(t) = \sum_n u_{kn}(t)\, e^{jn\gamma} \tag{39}$$

K. Torque equation — complete derivation and sign correction

Torque equation according to W. Leonhard:

$$T_M(t) = \frac{M}{3\pi} \int_0^{2\pi} \frac{i_S\, i_R^*\, e^{-j\varepsilon(t)} - i_S^*\, i_R\, e^{j\varepsilon(t)}}{2j}\, d\beta \tag{40}$$

The integrand does not depend on β, so the integral is resolved as the integrand multiplied by 2π:

$$T_M(t) = \tfrac{2}{3}\, M \cdot \frac{i_S\, i_R^*\, e^{-j\varepsilon} - i_S^*\, i_R\, e^{j\varepsilon}}{2j} \tag{41}$$

We define $W \equiv i_S(t) \cdot i_R^*(t) \cdot e^{-j\varepsilon(t)}$ (where $i_S \equiv i_0$ and $i_R \equiv i_1$ in the general notation of this section), with which $W^* = i_S^*(t) \cdot i_R(t) \cdot e^{j\varepsilon(t)}$, and since $(W - W^*)/(2j) = \mathrm{Im}\{W\}$:

$$W \equiv i_S(t)\, i_R^*(t)\, e^{-j\varepsilon(t)}\ , \qquad W^* = i_S^*(t)\, i_R(t)\, e^{j\varepsilon(t)} \tag{41a}$$

$$\frac{W - W^*}{2j} = \mathrm{Im}\{W\} \tag{41b}$$

the intermediate form given in the original paper is obtained:

$$T_M(t) = \tfrac{2}{3}\, M\, \mathrm{Im}\{i_S(t)\, [i_R(t)\, e^{j\varepsilon(t)}]^*\} \tag{42}$$

Identity (9) is now applied, defining $A = i_S(t)$ and $B = i_R(t) \cdot e^{j\varepsilon(t)}$. First we verify that $A \cdot B^*$ coincides exactly with W:

$$AB^* = i_S(t)\, [i_R(t)\, e^{j\varepsilon(t)}]^* = W \qquad \checkmark \tag{42a}$$

and since $\mathrm{Im}(AB^*)$ is precisely the left-hand side of identity (9):

$$\mathrm{Im}(AB^*) = \frac{AB^* - A^*B}{2j} = \vec{A} \times \vec{B} \tag{42b}$$

we obtain:

$$T_M(t) = \tfrac{2}{3}\, M \left[\vec{\imath}_S(t) \times \vec{\imath}_R(t)\, e^{+j\varepsilon(t)}\right] \tag{43}$$

Sign correction. The original text of this publication presents the final equation as:

$$\vec{T}_M(t) \textbf{ (original text)}: \quad \tfrac{2}{3}\, M\Big[\vec{\imath}_S(t) \times \vec{\imath}_R(t)\, e^{-j\varepsilon(t)}\Big] \tag{43a}$$

that is, with exponent $-\varepsilon(t)$. Mechanically applying the only tool the paper itself offers for this step —identity (9)— the sign that results from the complete derivation is $+\varepsilon(t)$, not $-\varepsilon(t)$. Since the derivation (40)–(43) is a closed, step-by-step verifiable algebraic chain starting from Leonhard's own equation, (43) is adopted here as the corrected form, and the discrepancy is attributed to a transcription error of the exponent's sign in the original document. This correction was additionally verified mechanically using the ACL2 theorem prover —independently of the algebraic development presented here— obtaining exactly the same sign $+\varepsilon(t)$ [3].

L. Final model of the asynchronous AC machine

Introducing equivalence (9), the final model of the asynchronous AC machine is:

$$\vec{u}_k(t) = R_k\, \vec{\imath}_k(t) + \dot{\vec{\psi}}_k(t) \tag{44}$$

$$J\, \dot{\vec{\omega}}_M(t) = \vec{T}_M(t) - \vec{T}_L(t) \tag{45}$$

$$\vec{T}_M(t) = \tfrac{2}{3}\, M\Big[\vec{\imath}_S(t) \times \vec{\imath}_R(t)\, e^{+j\varepsilon(t)}\Big] \tag{46}$$

Equation (46) reproduces exactly (43) —both already with the corrected sign, unlike the equation with the original text shown above only for comparison purposes— and closes the model: J is the rotor's moment of inertia, $\vec{T}_L(t)$ the external load torque.

IV. Conclusion

The development of the asynchronous AC machine has been presented starting from the complex-domain model, from which a vectorial model was obtained that reveals the nature of the motor torque as a cross vector product of the rotating stator and rotor current vectors. Relative to the original publication, this version completes three derivations that were left implicit there —the self and mutual inductances, the three-phase collapse of the flux linkage, and the complete chain leading to the torque equation— and corrects, with explicit algebraic justification and independent mechanical verification via ACL2 [3], the sign of the exponent in the final torque equation. The use of this representation is proposed for educational purposes, where the explanation of current interaction is made graphical through vector operations, and it can also be evaluated numerically, being equivalent to the original complex model from which it was derived.

Future Work

The symbolic verification originally planned using a computer-algebra system such as Maxima was replaced by a more rigorous formal verification: the algebraic core of the model —the identities (6)–(9), the three-phase collapse lemma of Section III.H, and the complete chain of the torque sign correction of Section III.K— was mechanically proved in ACL2, including a concrete instantiation with real $\gamma = 2\pi/3$ via ACL2(r) [3]. It remains to formalize in the same way the integration of the self and mutual inductances (Section III.F), for which the real integral-calculus library of ACL2(r) (`nonstd/integrals`) has already been identified as the formalization path, and the exact antiderivative needed was obtained by hand; the mechanization of the differentiation step remains. Numerical verification of the complete model —including the corrected torque sign— via Scilab simulation also remains pending.

V. Cross-reference to the ACL2 books

The ACL2 books are in the ACL2 community books, directory `books/workshops/2026/cayuela-ac-machine/` [3]. The first table gives, for each theorem, the equation(s) of this appendix that it states or proves. Theorems that only support a proof (arithmetic lemmas) are not listed.

Theorem	Equation(s)
00-complex-vectors.lisp	
`eq6-dot-product-real-part`, `eq6-dot-product-imag-part`	(6), (7)
`eq8-9-cross-product-real-part`, `eq8-9-cross-product-imag-part`	(8), (9)
`eq9-im-ab-star-is-crossp`	(9), (42b)
01-euler-identities.lisp	
`pythagorean-cos-sin`, `cosine-even`, `sine-odd`, `cosine-difference`, `sine-difference`, `acl2-numberp-cosine`, `acl2-numberp-sine`	assumed properties of cos and sin (`encapsulate`), used in (10)–(12) and (34g)
`eq10-euler-cosine`	(10)
`eq12-cosine-difference-is-dotp`	(12), through (11)
02-roots-of-unity.lisp	
`eq-roots-of-unity-sum-zero`	(19): $1 + e^{j\gamma} + e^{j2\gamma} = 0$
`eq-roots-of-unity-double-angle-sum-zero`	(34e)
03-zeta-deduction.lisp	
`eq16-28-re-ik-balanced`	(23)
`eq16-28-imz-ik`	(21)
`eq-ik-magnitude-squared`	(27)
04-collapse-lemma.lisp	
`eq33-35-three-phase-collapse-lemma`	(34a); its proof is (34b)–(34e), and its applications (34f), (34h)
`eq-mutual-phase-shift`	(34g)
05-torque-sign-correction.lisp	
`eq-W-conjugate`	(41a): W^*
`eq-w-minus-wconj-is-2j-im`	(41b)
`eq-ab-star-is-w`	(42a)
`eq-im-w-is-crossp`	(42b), (43)
`torque-exponent-sign-is-plus`	(43), against (43a)
`sign-matters-nonvacuous`	(43) and (43a) differ: the sign matters
06-generic-omega-collapse.lisp	
`trinary-collapse-lemma` (and its supporting lemmas)	(34a), for any ω with $\omega^2 + \omega + 1 = 0$
06-generic-omega-collapse-acl2r.lisp	
`cos-gamma-val`, `sin-gamma-val`, `omega-real-closed-form`, `omega-real-satisfies-cubic`	(19): $\omega = e^{j2\pi/3}$
`trinary-collapse-CONCRETE`	(34a), with $\omega = e^{j2\pi/3}$
07-inductance-note.lisp	
(no theorems)	(29)–(31): the integral (29e) is not mechanized
08-inductance-integral.lisp	
`cos-product-to-sum`	(29e): product-to-sum identity
`inductance-integrand-as-sum`	(29e): integrand as a sum

Theorem	Equation(s)
`sin-2x-minus-c-deriv`	derivative of the antiderivative of (29e)
09-nonvacuity-acl2r.lisp	
`sqrt-3-squared-nv`, `realp-sqrt-3`, `s3-hypothesis-satisfiable`	(19): $\sqrt{3}$ is real with square 3
`eq33-35-collapse-concrete`	(34a)
`eq-ik-magnitude-squared-concrete`	(27)
`sign-matters-concrete`	(43), (43a)
`acl2-trig-pythagorean`, `acl2-trig-cosine-even`, `acl2-trig-sine-odd`, `acl2-trig-cosine-difference`, `acl2-trig-sine-difference`, `acl2-trig-numberp-cosine`, `acl2-trig-numberp-sine`	the constraints of the `encapsulate`, proved for `acl2-cosine` and `acl2-sine`
`eq10-euler-cosine-r`, `eq12-cosine-difference-is-dotp-r`	(10), (12)
`eq-mutual-phase-shift-r`	(34g)
`eq-W-conjugate-r`	(41a)
`eq-im-w-is-crossp-r`, `torque-exponent-sign-is-plus-r`	(42b), (43)

Some comments inside the books refer to "images". The source document of this appendix was written with every equation and figure as an image, numbered in order of appearance: figure 1 is image 1, so an equation has image number equal to its original position plus the figures before it. The second table translates those comments.

Comment in the book	Image	Equation(s)
`02-roots-of-unity`: "Image 20"	20	(19), the two complex roots
`04-collapse-lemma`: "images 56-60"	56–60	(34a)–(34e), lemma and proof
`02-roots-of-unity`: "images 58-60"	58–60	(34c)–(34e)
`02-roots-of-unity`: "Image 60"	60	(34e), sum of the roots with the doubled angle
`04-collapse-lemma`: "image 62"	62	(34g), phase shift
`05-torque-sign-correction`: "image 74"	74	(42a), $AB^* = W$ with the check mark

References

[1] W. Leonhard, *Control of Electrical Drives*, 2nd and 3rd ed. Springer 1996–2001, Chapter 10, Symmetrical Three-Phase AC Machines.

[2] T. Needham, *Visual Complex Analysis*, 1st ed. Oxford: Clarendon Press, 2000, Ch. 1, Part III, Section 6, Vectorial Operations.

[3] P. O. Cayuela, "Mechanized Verification of Complex-to-Vector Derivations in a Three-Phase AC Machine Model," 20th International Workshop on the ACL2 Theorem Prover and Its Applications (ACL2-2026), EPTCS, to appear. The books are in the ACL2 community books, `[books]/workshops/2026/cayuela-ac-machine/`; development version: https://github.com/pcayuela/ACL2-ACmotormodel-check